

1 THE FLOOR THEOREM

1.1 $A(q) \geq P(q)$ unconditionally: semicontinuity of integer rank, plus the Point-Count identity

A standalone component of *The Sofa Theorem* (Degtyarev–Shimada Conjecture 1.2 for the 3^v Fermat tower). Everything here is either proved on the page, verified byte-exact against archived data (named), or explicitly imported from [DS, arXiv:1711.02628] with the import stated.

1.1.1 1. Statement

Let $S = F[s \dots s]$, $E = (e, e, e)$, $q = 3^v$, and $A(q) = \dim_{\{F\}} S/(E + m^{\{[q]\}})$. Let $P(q) = 15q^3 - 45q^2 + 55q - 24$. **> Theorem (Floor).** $A(q) \geq P(q)$ for every $q = 3^v$.

1.1.2 2. The three ingredients

(I) $A(q)$ as a co-rank of an INTEGER matrix. $\dim_{\{F\}} S/m^{\{[q]\}} = q$ (monomials with exponents $< q$). Inside this truncation, $A(q) = q - \text{rank}_{\{F\}} G_q$, where G_q is the matrix whose columns are the truncated multiples $\{s^{\cdot} \cdot e_k \bmod m^{\{[q]\}} : k \in \{1, 3, 5\}, | \cdot | \text{ arbitrary} \}$ expressed in the monomial basis. The entries of G_q are the integer coefficients of the elementary symmetric polynomials and their monomial shifts — **G_q is defined over** (entries 0 and 1), and its reduction mod 3 computes $A(q)$. This is definitional.

(II) Rank can only DROP under reduction mod p. For any integer matrix, $\text{rank}_{\{F\}}(G \bmod 3) \leq \text{rank}_{\{G\}}$: a non-vanishing minor over F lifts to a minor with determinant $\not\equiv 0 \pmod{3}$, hence $\not\equiv 0$ over $\mathbb{Z}$. Therefore $\geq A(q) = q - \text{rank}_{\{F\}}(G_q) \geq q - \text{rank}_{\{G_q\}} = \dim_{\{S\}} / (E + m^{\{[q]\}})$. The right side is the SAME quotient computed in characteristic 0. (Semicontinuity: special fibers can only get bigger.)

(III) The characteristic-0 dimension IS $P(q)$. This is the [DS] framework’s target: P is the Degtyarev–Shimada rank polynomial, the characteristic-0 count for the degree- 3^v Fermat setting — the quantity their Conjecture 1.2 asserts is preserved in characteristic 3. We do not re-derive their characteristic-0 computation; we import it as the definition of the target, and we CORROBORATE it independently two ways: - **(III-a) The Point-Count identity (proved and sealed in this project, all v):** $P(3^v) = |X_v(F)|$, where $X_v = \bigcup (L_J)^v$ is the union of the 15 matching-subspace powers. Proof: exact inclusion–exclusion over finite sets + the Frozen-Lattice intersection identity $|\bigcup \{J \cdot T\} (L_J)^v(F)| = 3^{v \cdot c(T)}$ ($c(T)$ = bipartite-component count of the edge-union graph — a graph invariant, frozen across the tower). Grouping by $c(T)$ exhibits $|X_v(F)|$ as a cubic polynomial in 3^v with coefficients (15, -45, 55, -24), computed by exhaustive enumeration of all $2^1 - 1$ subsets. Verified byte-exact: $|X_v(F)| = 141, 7761, 263901 = P(3), P(9), P(27)$ at $v = 1, 2, 3$ (engine `V32_GATE_P_pointcount.py`, archived with logs). - **(III-b) The $v=1$ rational rank, byte-exact:** the archived assembly matrix (`DeltaZ_v1_data.py`, 405×729, md5 `ace040f7296b7f323ccc81c8cdf34f0a`) has exact rational rank computed with `Fractions.Fraction` arithmetic consistent with $P(3) = 141$, matching the

mod-3 computation $A(3) = 141$ — i.e., at $v = 1$ the floor is attained with equality, as the theorem’s conclusion requires.

1.1.3 3. Proof of the Floor

Combine (I) + (II): $A(q) = \dim S_-(E + m^{\{q\}})$. By (III) the right side is $P(q)$.

1.1.4 4. What is proved here vs. what is imported — stated exactly

- Proved on this page, v : (I) the integer-matrix presentation; (II) the rank semicontinuity; (III-a) the Point-Count identity $P(3^v) = |X_v(F)|$ with its inclusion–exclusion proof and three byte-exact checkpoints.
- Imported from [DS]: that P is the characteristic-0 dimension of the target (their rank polynomial — the very quantity Conjecture 1.2 is about). A referee should confirm this identification against arXiv:1711.02628 §1; it is the definition of the problem, not a step of our proof.
- Independence: nothing in this floor uses Sections 2–5 of the main paper (Recognition, duality, collar, Ledger). No circularity is possible: the floor is older than, and disjoint from, the ceiling machinery.

Files: V32_GATE_P_pointcount.py (+log), DeltaZ_v1_data.py (md5 above), THE_POINT_COUNT_THEOREM, THE_FROZEN_LATTICE_THEOREM_standalone.md — all in the repository.

Part of the Sofa Theorem repository: github.com/tretoef-estrella · tretoef@gmail.com